

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

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PART 1 - Problem 2 - State Movement of an Autonomous LTI System

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC]

Consider the LTI autonomous system

$$\dot{x}(t) = A x(t)$$

where the matrix A is

$$A = \begin{bmatrix} 0 & -2 & 0 \\ -2 & -3 & 1 \\ 0 & 0 & -5 \end{bmatrix}$$

Assume as the **initial state** the following, generic, state x_0 :

$$x(0) = x_0 = \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix}, \quad x_a, x_b, x_c \in \mathbb{R}$$

The (free) state trajectory starting from $x(0)$ at the time instant $t = 0$ can be calculated using the Laplace transform

$$x(t) = \mathcal{L}^{-1} \left[(sI - A)^{-1} x(0) \right] = e^{A t} x(0), \quad \forall t \geq 0$$

Your code should:

- Declare the **symbolic** variables, named **x_a**, **x_b** and **x_c**, corresponding to the components of the initial state $x_0 = [x_a \ x_b \ x_c]^T$.
- Declare the **symbolic** variable **matA** corresponding to the matrix A .
- Compute the expression of the matrix exponential $e^{A t}$ and assign the result to the **symbolic** variable **expAt**.
- Determine the analytic expression of the state movement $x(t)$, with initial state $x(0) = x_0$. Assign the obtained expression to the **symbolic** variable **x_t**.
- Verify that the analytic expression of **x_t** contains the terms e^t , e^{-4t} , and e^{-5t} .
- Consider now the new initial state

$$x(0) = \bar{x} = \begin{bmatrix} x_a \\ +\frac{5}{2}x_a \\ -3x_a \end{bmatrix}, \quad x_a \in \mathbb{R}$$

i.e., you force the state components x_b and x_c to be equal, respectively, to

$$x_b = +\frac{5}{2}x_a, \quad x_c = -3x_a, \quad x_a \in \mathbb{R}$$

- Determine the analytic expression of the new state movement $x(t)$, with initial state $x(0) = \bar{x}$. Assign the obtained expression to the **symbolic** variable **xNEW_t**.
- **HINT**: You may determine the new analytical expression of the state movement using the general formula $x(t) = \mathcal{L}^{-1}[(sI - A)^{-1}x(0)] = e^{At}x(0)$, $\forall t \geq 0$ (already used above). Alternatively, exploiting the MATLAB command **subs**, you may use the already computed expression **x_t** and substitute the new expressions in place of the generic initial conditions **x_b** and **x_c**.
- Verify that the analytic expression of the state movement **xNEW_t** contains ONLY the term e^{-5t}

Assignments

1. Verify the existence of **x_a**, **x_b** and **x_c**..
2. Check if **expAt** has the desired expression.
3. Check if **x_t** is the correct state movement.
4. Check if **xNEW_t** is the correct state movement.

Solution

```
syms t s
syms x_a x_b x_c
assume(t>=0)

% the initial condition
x0 = [x_a; x_b; x_c];
% the matrix A
matA = sym([0 -2 0; -2 -3 1; 0 0 -5]);
% the symbolic expression (sI-A)^-1
sI_A = s*eye(size(matA)) - matA;
sI_A_INV = simplify(inv(sI_A));

% the matrix exponential
expAt = simplify(collect(ilaplace(sI_A_INV), [exp(t), exp(-4*t), exp(-5*t)]))

expAt =
```

$$\begin{pmatrix} \frac{4e^t}{5} + \frac{e^{-4t}}{5} & \sigma_1 & -\frac{e^t}{15} + \sigma_2 - \frac{e^{-5t}}{3} \\ \sigma_1 & \frac{e^t}{5} + \frac{4e^{-4t}}{5} & \frac{e^{-5t}(e^{6t} + 24e^t - 25)}{30} \\ 0 & 0 & e^{-5t} \end{pmatrix}$$

where

$$\sigma_1 = -\frac{2e^t}{5} + \sigma_2$$

$$\sigma_2 = \frac{2e^{-4t}}{5}$$

```
% the free state trajectoty starting
% from x0 at the time instant t=0
X_s = sI_A_INV * x0;
x_t = simplify(collect(ilaplace(X_s, s, t), [exp(t), exp(-4*t), exp(-5*t)]))
```

$$x_t = \begin{pmatrix} e^{-4t} \left(\frac{x_a}{5} + \frac{2x_b}{5} + \frac{2x_c}{5} \right) - e^t \left(\frac{2x_b}{5} - \frac{4x_a}{5} + \frac{x_c}{15} \right) - \frac{\sigma_1}{3} \\ e^t \left(\frac{x_b}{5} - \frac{2x_a}{5} + \frac{x_c}{30} \right) - \frac{5x_c e^{-5t}}{6} + e^{-4t} \left(\frac{2x_a}{5} + \frac{4x_b}{5} + \frac{4x_c}{5} \right) \\ \sigma_1 \end{pmatrix}$$

where

$$\sigma_1 = x_c e^{-5t}$$

```
% state evolution with IC = barX
xNEW_t = collect(subs(x_t, [x_c, x_b], [-3*x_a, 5*x_a/2]))
```

$$xNEW_t = \begin{pmatrix} e^{-5t} x_a \\ \frac{5e^{-5t}}{2} x_a \\ (-3e^{-5t}) x_a \end{pmatrix}$$